

TLU Plus File

Monday, September 21, 2026 11:48 AM

VLSI Physical Design Learning Series

Parasitic Extraction & STA

Example of a TLUPlus File — Part 1

File identity, ASCII header, units, technology metadata & interconnect structure



Important: TLUPlus is a binary parasitic-model file with an ASCII header. The detailed model data (R/C tables) is not shown here. The content shown here is an EDUCATIONAL / LOGICAL REPRESENTATION to explain the structure and information. Actual syntax may vary.

1 Illustrative TLUPlus Header / Logical View

```

1  # =====
2  # TLUPlus technology model - logical representation
3  # (Illustrative example - not actual file syntax)
4  # =====
5
6  TECHNOLOGY      = "Example_7nm"      # technology/node
7  PROCESS         = "typical"          # process corner
8  MODEL           = "TLUPlus"          # model type
9  VERSION         = "example"          # model version
10
11 UNITS_LENGTH     = "micron"           # length unit
12 UNITS_RESISTANCE = "ohm"              # resistance unit
13 UNITS_CAPACITANCE = "ff"              # capacitance unit
14 TEMPERATURE     = 25                  # reference temp (°C)
15 CORNER           = "RC_WORST"        # extraction corner
16
17 LAYERS           = ( M1 M2 M3 M4 M5 ) # metal layers
18 VIAS             = ( V12 V23 V34 V45 ) # via layers
19
20 # -----
21 # binary model data (R/C tables, advanced models, etc.)
22 # -----
23 ... BINARY MODEL DATA (NOT HUMAN READABLE) ...

```

Note: Real TLUPlus files are binary. The ASCII header identifies the model, units, layers, process corner and other context information.

2 What is TLUPlus ?

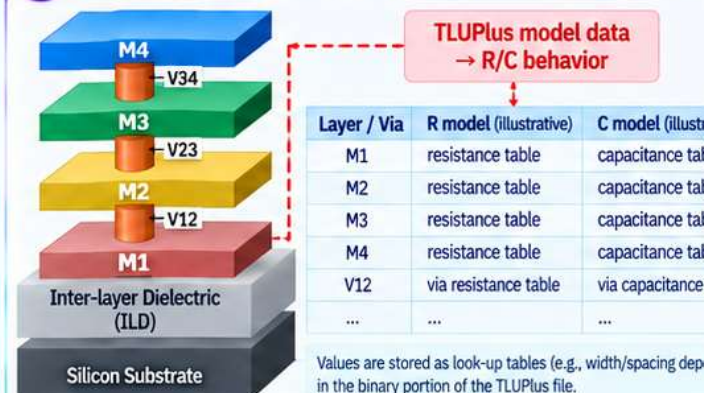
- TLUPlus provides resistance (R) and capacitance (C) look-up data for interconnects (metals and vias) used in parasitic estimation and extraction.
- It models advanced process effects (e.g., layer-dependent R/C, coupling, dielectric effects) for accurate parasitics.
- Widely used in Synopsys flows (e.g., IC Compiler, PrimeTime).
- Helps improve correlation with signoff/extracted results.
- In Synopsys flows, TLUPlus files are specified using:

```

set_tlu_plus_files \
-max_tluplus <file> \
-min_tluplus <file> \
[-tech2itf_map <map_file>]

```

4 Interconnect Structure and TLUPlus Modeling Illustration



Part 1 Takeaway:

- TLUPlus is not a DEF/LEF layout file; it is technology-dependent interconnect RC model data.
- Real files are commonly binary with an ASCII header.
- The next parts will show logical representations of layer resistance tables, capacitance tables, mapping/cornering.

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TLUPlus (Parasitic Extraction)

Example of a TLUPlus File — Part 2

Layer definitions, thickness, width/spacing constraints and layer parameters

1 TLUPlus Content (Part 2) — Layer and Via Definitions (Example)

```

1  # =====
2  # LAYER DEFINITIONS
3  # (Metal layers with thickness, width/spacing range, etc.)
4  # =====
5  LAYER M1
6      TYPE      = "ROUTING"
7      THICKNESS  = 0.040      # microns (40 nm)
8      MIN_WIDTH  = 0.040
9      MAX_WIDTH  = 10.000
10     MIN_SPACING = 0.040
11     MAX_SPACING = 10.000
12     R_TABLE     = "M1_R_TBL"
13     C_TABLE     = "M1_C_TBL"
14     DIRECTION   = "HORIZONTAL"
15     HEIGHT      = 0.10      # distance from previous layer (ILD)

```

2 Metal Layer Parameters

Parameter	Meaning
TYPE	Layer type (routing, core, etc.)
THICKNESS	Physical metal thickness
MIN_WIDTH / MAX_WIDTH	Width range for which the model is valid (μm)
MIN_SPACING / MAX_SPACING	Spacing range for coupling
R_TABLE	Name of resistance look-up table
C_TABLE	Name of capacitance look-up table

```

15
16 LAYER M2
17 TYPE = "ROUTING"
17 THICKNESS = 0.050
18 MIN_WIDTH = 0.050
19 MAX_WIDTH = 10.000
20 MIN_SPACING = 0.050
21 MAX_SPACING = 10.000
22 R_TABLE = "M2_R_TBL"
23 C_TABLE = "M2_C_TBL"
24 DIRECTION = "VERTICAL"
25 HEIGHT = 0.12
26 ...
27
28 #=====
28 # VIA DEFINITIONS
28 #=====
29
30 VIA V12
31 LAYER1 = "M1"
32 LAYER2 = "M2"
33 RES_TABLE = "V12_R_TBL"
34 CAP_TABLE = "V12_C_TBL"
35 MIN_SIZE = 0.040
36 MAX_SIZE = 1.000
37 CUT_SPACING = 0.040
38 CUT_ROWS = 1
39 CUT_COLS = 1
40 ...

```

DIRECTION	Preferred routing direction
HEIGHT	Inter-layer height (dist previous layer, μm)

These parameters define the physical and electrical properties of the layer. The actual R/C values are stored in separate tables.

3 Via Layer Parameters (Example)

Parameter	Meaning
LAYER1 / LAYER2	Layers connected by this via
RES_TABLE	Via resistance lookup table
CAP_TABLE	Via capacitance lookup table
MIN_SIZE / MAX_SIZE	Valid via size range (μm)
CUT_SPACING	Spacing between via cuts (μm)
CUT_ROWS / CUT_COLS	Number of cuts in via array

Via resistance and capacitance are also modeled using lookup tables. Different via types (e.g., V12, V23) will have their own tables.



Part 2 Takeaways:

- Each metal layer and via layer is defined with physical and electrical parameters.
- Lookup tables (R/C) are referenced for accurate parasitic modeling.
- These definitions establish the valid geometry range for extraction.

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TLUPlus (Parasitic Extraction)

Example of a TLUPlus File — Part 1: R/C Tables, Coupling Models, Temperature Corners & Advanced Options

1 TLUPlus Content (Part 3) — R, C and Coupling Tables (Example)

```

1 # =====
2 # RESISTANCE TABLE (R) — Layer M1 (Example)
3 # Values: ohms per micron (scaled by width and length)
4 # =====
5 LAYER M1
6 R_TABLE
7 # W (\(\mu\text{m}\)) -> 0.04 0.08 0.16 0.32 0.64 1.28
8 L=0.04 0.152 0.098 0.071 0.052 0.041 0.033
9 L=0.08 0.160 0.104 0.075 0.055 0.044 0.036
10 L=0.16 0.176 0.113 0.081 0.060 0.047 0.038
11 L=0.32 0.198 0.126 0.090 0.066 0.051 0.041
12 END_TABLE
13
14 # =====
15 # CAPACITANCE TABLE (C) — Layer M1 to Substrate (Example)
16 # Values: fF per micron (dependent on width and spacing)
17 # =====
18 LAYER M1
19 C_TABLE
20 # W\ S (\(\mu\text{m}\)) -> 0.04 0.08 0.16 0.32 0.64 1.28
21 0.04 0.215 0.182 0.150 0.122 0.098 0.081
22 0.08 0.228 0.195 0.161 0.132 0.107 0.088
23 0.16 0.246 0.210 0.174 0.142 0.115 0.095
24 0.32 0.268 0.229 0.190 0.154 0.124 0.101
25 END_TABLE
26
27 # =====
28 # COUPLING CAPACITANCE TABLE (Cc) — Layer M1 to M1 (Example)
29 # Values: fF per micron (depending on spacing and length)
30 # =====

```

2 Temperature Corners and Process Variations

Parameter	Example Entry			
TEMPERATURE	25	125		List of temperatures (in $^{\circ}\text{C}$)
PROCESS	TYP	WORST	BEST	Process variation corners
R_SCALE	1.0	1.15	0.85	Scaling factor for resistance (per corner)
C_SCALE	1.0	1.10	0.90	Scaling factor for capacitance (per corner)
CONSISTENT_UNITS	YES			Ensures consistent units across all tables

Different process and temperature corners allow for accurate modeling of parasitics for multiple PVT conditions (e.g., typical, worst-case, best-case).

3 Advanced Options in TLUPLUS

Option	Example Entry			
DIELECTRIC_MODEL	CONSTANT_K	/	FIELD_SOLVER	Type of dielectric model
COUPLING_MODEL	SIMPLE	/	ADVANCED	Fring field model
LATERAL_COUPLING	ENABLED			Model lateral coupling between layers
DIAGONAL_COUPLING	ENABLED			Model diagonal coupling between layers
EDGE_CAP_MODEL	ENABLED			Included edge capacitance for a via

32	LAYER	M1	M1				
34	CC_TABLE						
33	# S (um)	->	0.04	0.08	0.16	0.32	0.64
38	L=0.04		0.118	0.074	0.045	0.026	0.015
35	L=0.08		0.122	0.078	0.047	0.027	0.016
35	L=0.16		0.129	0.082	0.050	0.029	0.017
37	L=0.32		0.138	0.087	0.053	0.031	0.018
38	END_TABLE						

FREQUENCY_DEPENDENT	NO	Freq extr
COMMENT	"Example 7nm TLUPlus"	User

These options control the accuracy and complexity of the extraction. Foundry PDKs typically provide recommended settings.



Part 3 Takeaways:

- R, C and coupling tables model the electrical behavior of metal layers and vias.
- Temperature and process corners allow variation-aware extraction.
- Advanced options (coupling, lateral, fringe, etc.) improve accuracy for signal integrity.

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TLUPlus (Parasitic Extraction)

Example of a TLUPlus File — Part 3: Advanced Sections: Cut/Spacing Tables, Thickness & Dielectric

1 Via Cut Resistance and Capacitance Tables (Example)

```

1 # =====
2 # VIA CUT TABLES (R and C)
3 # Example for Via type V12 (M1 to M2)
3 # Units: Resistance in ohms, Capacitance in fF
4 # =====
5 VIA      V12
6 RES_TABLE
7 # Rows \ Cols   1       2       3       4
8   1       0.120    0.090    0.075    0.068
9   2       0.110    0.082    0.070    0.062
10  3       0.105    0.078    0.066    0.058
11  4       0.100    0.074    0.061    0.053
12 END_TABLE
13
14 CAP_TABLE
15 # Rows \ Cols   1       2       3       4
16   1       0.35    0.62    0.88    1.10
17   2       0.50    0.82    1.15    1.42
18   3       0.78    1.20    1.65    2.05
19   4       1.05    1.60    2.10    2.60
20 END_TABLE

```

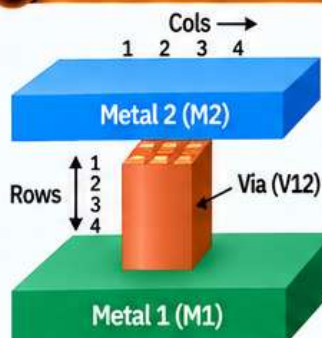
2 Spacing Dependent Capacitance (Fringe/Side/Overlap)

```

1 # =====
2 # SPACING DEPENDENT CAPACITANCE TABLE
3 # Example for M1 (fringing and lateral coupling)
3 # Units: fF/um
4 # =====
5 LAYER      M1
6 SPACING_TABLE
7 # S (um)    C_fringe    C_side    C_overlap
8   0.04      0.115      0.102      0.085
19  0.08      0.162      0.078      0.066
11  0.16      0.120      0.054      0.046
12  0.32      0.088      0.038      0.032
13  0.64      0.065      0.025      0.021
14  1.28      0.050      0.018      0.015
15 END_TABLE
16
17 # Width dependent fringe capacitance (optional)
18 WIDTH_FRINGE_TABLE
19 # W (um)    C_fringe (fF/um)
20  0.04      0.320
21  0.08      0.280
22  0.16      0.245
23  0.32      0.210
24  0.64      0.180
25  1.28      0.150
26 END_TABLE

```

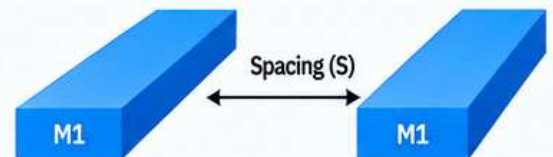
4 Via Geometry Illustration



Via Cut Table Representation

- R/C values are provided for different number of via cuts (rows x cols).
- More cuts → lower resistance, higher capacitance.
- Used during extraction to model via parasitics accurately.

5 Spacing Dependent Capacitance Illustration



- Capacitance between adjacent wires depends on spacing (S).
- TLUPlus provides fringe, side and overlap capacitance values for different spacing values.
- These tables help Quantus model coupling accurately.



Part 4 Takeaways:

- Via cut tables, spacing tables and dielectric information provide detailed parasitic modeling.
- These parameters are technology specific and highly influence RC extraction accuracy.
- Proper modeling of vias, dielectrics and spacing leads to better STA correlation and timing closure.

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Example of a TLUPlus File — Part 4: Dielectric & Thickness

TLUPlus (Parasitic Extraction) Layer mapping, corner usage, tool setup, validation & complete TLUPlus

1 TLUPlus Setup / Corner Association (Illustrative)

```

1 # =====
2 # TLUPlus setup in Synopsys flow (Example)
3 # (Command syntax and options are tool/version dependent)
4 # =====
5
6 # Set max and min TLUPlus files and technology-layer map
7 set_tlu_plus_files \
8   -max_tlu_plus rcworst.tluplus \ # worst RC (max R, max C)
9   -min_tlu_plus rcbest.tluplus \  # best RC (min R, min C)
10  -tech2itf_map tech2itf.map      # layer name mapping file
11
12 # (Optional) For a typical corner
13 set_tlu_plus_files -tlu_plus rctyp.tluplus
14
15 # View the current TLUPlus setup
16 report_tlu_plus_files
17
18 
```

Note: This is an ILLUSTRATIVE / LOGICAL REPRESENTATION.
Exact syntax, options and file names vary by Synopsys tool, release and foundry.

2 Technology-Layer Mapping

Design layer names (from LEF/DEF) are mapped to technology layer names used in the TLUPlus file using a tech2itf.map file.

Design / ITF Layer (name in LEF/DEF)		Technology Layer (name in TLUPlus / ITF)
M1	↔	METAL1
M2	↔	METAL2
M3	↔	METAL3
M4	↔	METAL4
V12	↔	VIA12
V23	↔	VIA23

Layer mapping is required when design layer names and technology layer names are different.
The tech2itf.map file provides this mapping.

3 RC Corner Usage

RC Corner	TLUPlus File
RC_WORST	max_tlu_plus (rcworst.tluplus)
RC_BEST	min_tlu_plus (rcbest.tluplus)
RC_TYP	typical model (rctyp.tluplus)

Exact setup/hold corner assignment and file names are methodology & tool dependent. Always follow your foundry's recommendations.

4 How the Tool Uses TLUPlus (During Extraction and STA)

Routed DEF
(Physical geometry)

Shapes, wires, vias, coordinates

+

LEF / ITF
(Layer information)

Layer names, widths, spacing, stack-up

+

TLUPlus
(R/C lookup models)

Resistance & capacitance tables for each layer/via

→

Quantus / QRC
(Parasitic Extraction)

Uses geometry + layer info + TLUPlus models to extract R & C

→

Extracted Parasitics
(SPEF / RCDB)

Distributed R and C with corner information

6 Complete Practical Flow (TLUPlus → Extraction → STA)

Foundry PDK
(ITF / tech files)

Technology models, layer stack, process data (e.g., .itf, .tf)

→

TLUPlus Models
(max / min / typ)

rcworst.tluplus
rcbest.tluplus
rctyp.tluplus

→

Layer Mapping
(tech2itf.map)

Layer name mapping (Design ↔ Technology)

→

P&R Tool
(ICC2 / Fusion Compiler)

Reads TLUPlus for extraction and analysis

→

RC Extraction
(Quantus / QRC)

Parasitic extraction using TLUPlus and routed design

Final Takeaway:

Correct TLUPlus + correct layer mapping + correct RC corner = reliable parasitic